

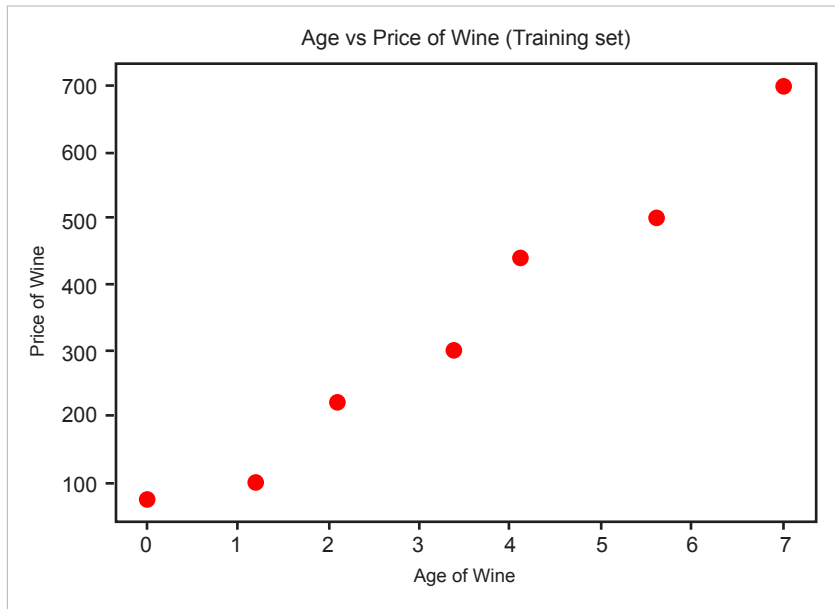


Figure 2: Data visualisation

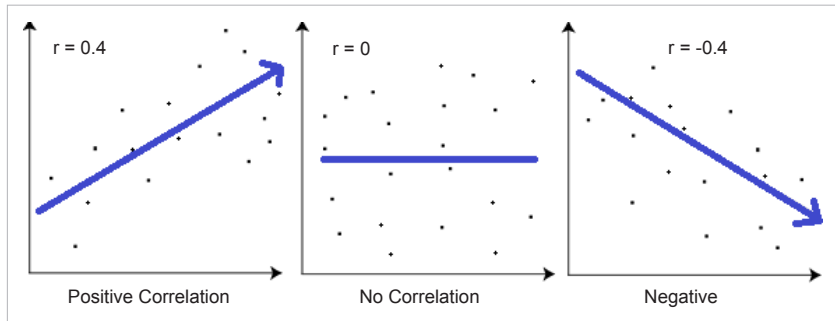


Figure 3: Correlation visualisation (Credits: <https://www.statisticshowto.com/probability-and-statistics/correlation-coefficient-formula/#Pearson>)

R can range from +1 to -1, but the further this figure is away from 0, the stronger is the correlation. '1' indicates a strong positive relationship and '-1' indicates a strong negative relationship. '0', on the other hand, indicates no relationship at all.

A graphical view of a correlation of -1, 0 and +1 is shown in Figure 3.

There are several types of correlation coefficients but the most popular is Pearson's Correlation. Its coefficient formula is seen in Figure 4.

The two variables are divided by the product of the standard deviation of each data sample. This is the normalisation of the covariance between the two variables to give an interpretable score.

Let's switch back to our scenario of age and price of wine to find out whether there is a correlation and, if yes, what is the value of R.

```
from scipy.stats import pearsonr
# calculate Pearson's correlation
corr, _ = pearsonr(list(X.flat), y)
print('Pearsons correlation: %.3f' % corr)
```

The output seen is:

Pearsons correlation: 0.985

	Age of wine	Price of wine	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x}) * (y - \bar{y})$
1	0	75	-3.34	-258.57	11.15	66859.18	864.37
2	1.2	100	-2.14	-233.57	4.57	54555.61	500.51
3	2.1	220	-1.24	-113.57	1.53	12898.47	141.15
4	3.4	300	0.06	-33.57	0	1127.04	-1.91
5	4.1	440	0.75	106.43	0.56	11327.04	80.58
6	5.6	500	2.26	166.43	5.11	27698.47	375.65
7	7	700	3.66	366.43	13.4	134269.9	1340.08
Sum	23.4	2335	0 (will always be zeros)	0 (will always be zero)	36.36 is Sxx	308735.71 is Syy	3300.42 is Sxy
Average	3.34 is $\bar{x}$	333.57 is $\bar{y}$					

Table 2